

Wenhao Zhao

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EDUCATION

Northeastern University

Process equipment and control engineering

Shenyang, LiaoNing

Sep. 2020 – Jul 2024

The Hong Kong University of Science and Technology(Guangzhou)

Robotics and Autonomous Systems

Guangzhou, Guangdong

Sep. 2024 – Present

TECHNICAL SKILLS

Languages: Python, C/C++, Matlab, C#, SQL, JavaScript, HTML/CSS

Frameworks: ROS1, ROS2, Node.js

Developer Tools: Solidworks, CAXA, Keil, Isaac lab, Airsim, Webots, Mujoco, VS Code, Visual Studio, PyCharm, Git

COMPETITION AWARDS

- The first prize in 20th National College Student Robot Competition ROBOCON2021.
- The second prize in the 2023 “Competitive World Cup” China College Student Computer Game Competition.
- The first prize in the Robot Competition for Undergraduate Students in Liaoning Province.
- The first prize in the Northeast Division of the 10th National College Student Optoelectronic Design Competition.
- The second prize of the 2021 TI Cup Liaoning College Student Electronic Design Competition.
- The first prize in the 2022 Northeastern University Innovation and Entrepreneurship Competition.

PUBLICATIONS

- L. Shi, Y. Xu, S. Wang, J. Huang, **W. Zhao**, Y. Jia, Z. Yan, W. Gu, and G. Zhou, “An Real-Sim-Real (RSR) Loop Framework for Generalizable Robotic Policy Transfer with Differentiable Simulation,” *arXiv preprint* arXiv:2503.10118, 2025. Accepted for publication in *IEEE Robotics and Automation Letters* (RA-L).
- Y. Liu, Y. He, **W. Zhao**, X. Mo, Y. Xu, H. Wei, M. Ma, H. Li, Y. Wu, Z. Dai, X. Zhou, and F. Gao, “TIO-Former: Ultra-Lightweight 6-Directional ToF-Inertial Odometry for Nano-UAVs via a Streaming Causal Transformer,” *arXiv preprint* arXiv:2609.17198, 2026. Submitted to *IEEE International Conference on Robotics and Automation* (ICRA); under review.
- Y. He, Y. Liu, **W. Zhao**, H. Lin, D. Zhang, M. Ma, F. Gao, H. Yu, Z. Dai, and Z. Ding, “OmniRisk: Omnidirectional Trajectory-Risk Learning for Agile Quadrotor Dynamic Avoidance,” *arXiv preprint* arXiv:2609.18191, 2026. Submitted to *IEEE International Conference on Robotics and Automation* (ICRA); under review.
- **W. Zhao**, Y. Liu, Y. He, Y. Ruan, W. He, E. Rong, Z. Dai, and X. Zhou, “FAST360: Full-Azimuth Depth Perception and Circular-Transformer Trajectory Planning for High-Speed Quadrotor Flight,” 2026. Manuscript in preparation for submission to *IEEE Robotics and Automation Letters* (RA-L).

RESEARCH EXPERIENCE DURING UNDERGRADUATE

Quadruped robots

Team Member, Action Lab, Northeastern University

- In the quadruped robot project based on model predictive control (MPC) and whole body motion control (WBC), participate in the webots simulation motion control to make the quadruped robot stand on three legs and bow down to accumulate strength, take off, retract the front legs, exert force on the back legs, and land smoothly in the jumping process, STM32 controls the position of the foot end of a single leg through positive kinematics, and the communication between STM32 and the host computer. Later I have utilized the Proximal Policy Optimization (PPO) algorithm for reinforcement learning on a quadruped robot, and employed the Isaac Gym for GPU-parallel training, completing the simulation process from standing to walking for the quadruped robot.

A robot that can flip a volleyball

Team Leader, Action Lab, Northeastern University

- We utilized the ROS2 framework to build a robot designed for playing volleyball, which is completely self-engineered from the mechanical structure to the hardware and software. I deployed reinforcement learning algorithms on it, enabling it to continuously juggle a volleyball on a disc approximately 300mm in diameter and keep the ball from falling for an extended period.

Momentum wheel system

Team Member, Action Lab, Northeastern University

- The low-pass filter is added by cascade PID, and the momentum wheel model is optimized by combining the existing anti-interference in China, that is, the angle of the equilibrium position is modified every once in a while, and the momentum wheel inverted pendulum model that can remain stationary at the equilibrium position and self-adjust the center of gravity is adjusted.

Hot rolling production scheduling system

Team Member, DAO Lab, Northeastern University

- In the steel hot rolling production scheduling optimization system based on heuristic search algorithm, the meta-heuristic algorithm was optimized by participating in the scheduling and scheduling system of the operation plan, and the data in the system were added, deleted, modified and checked by using SQL server and C# and displayed in the WinForm interface.

Inpainting

Responsible for code writing and debugging, Northeastern University

- It is mainly through the construction of a multi-stage generative adversarial network and the addition of GAM's global attention mechanism GAM, as well as U-net hopping connection. The optimization of the face restoration process was realized, and it proved to have stronger robustness and repair ability in the test sets CelebA and CelebA-HQ.

INTERNSHIP EXPERIENCE

Shenzhen Dongyu Technology Co., Ltd | *Software & Embedded Development*

Jan 2022 – Mar 2023

- The corresponding software was developed to realize remote control and management of the Bluetooth devices designed by the team, and the information and file transmission between Bluetooth groups that do not support Bluetooth networking were initially realized through priority search traversal based on signal strength.

HKUST(GZ) | *Research Assistant in Smart Manufacturing Thrust*

May 2022 – Mar 2023

- The project team participated in the research and development of the GB-X project, mainly responsible for building the back-end data management platform used by the project on the server, and realizing the connection between the R&D equipment and the back-end data through the communication protocol, and participating in the development of some front-end administrators using some functions of the platform.

Dongguan Benmo Technology Co., Ltd | *Algorithm Research & Development*

Sep 2025 – Nov 2025

- In this internship project, I mainly focus on training the TITA legged robot in Isaac Lab to generalize to rough terrain, enabling it to maintain a stable posture on various uneven surfaces and when going up and down stairs. I then build a MuJoCo-based simulation environment in Docker, use the ROS 2 communication framework, and deploy this policy in the MuJoCo environment. By introducing the ROBOCON 2026 competition arena into the simulation, I aim to organize and open-source a reinforcement learning baseline for the TITA platform to support the ROBOCON organizers.

Weifen Zhifei Co., Ltd. | *Algorithm Research & Development*

Nov 2025 – Nov 2026

- In this project, I was mainly responsible for conducting real-world flight experiments for obstacle avoidance using monocular depth estimation. I used an Insta360 camera to perform panoramic image-based depth estimation for the UAV. In the AirSim simulator, the UAV achieved a maximum flight speed of 30 m/s. In real-world flight tests, it reached a maximum speed of 20 m/s, representing the current state of the art for RGB-perception-based UAV obstacle avoidance. The panoramic camera also provided omnidirectional perception and dynamic obstacle avoidance capabilities.

MPHIL PROJECTS IN HKUST(GZ)

Aerial manipulation | *Hybrid control of the drone and robotic arm*

Sep 2024 – Now

- Considering that most current drones are mainly used for detection and reconstruction tasks and lack physical interaction with the environment, their application scenarios are significantly limited. This project aims to develop a drone system equipped with a mechanical gripper that can perform obstacle avoidance while stably grasping stationary or moving targets on or near the ground. I am not only involved in designing the drone's mechanical structure and gripper, but also primarily responsible for the coordinated control of the drone and the gripper. By using traditional optimal control and deep reinforcement learning (DRL) methods, we seek to achieve generalized grasping of diverse objects and improve the drone's attitude and flight stability after completing the grasp.

Tail Sitter | *Control of a Tail-Sitter VTOL UAV Based on Transformer*

May 2025 – Nov 2025

- Tail-sitter vertical takeoff and landing (VTOL) unmanned aerial vehicles (UAVs) have the capability of hovering and performing efficient level flight with compact mechanical structures. We present a unified controller design for such UAVs, based on transformer. An advantage of this design method is that the various flight modes (i.e., hovering, transition and level flight) of a VTOL UAV are controlled in a unified manner, as opposed to treating them separately and in the runtime switching one from another. We employ a transformer network to approximate the behavior of the nonlinear solver, which suffers from high computational complexity. The proposed transformer has negligible approximation errors, and can be implemented in realtime (e.g., 20 Hz). Moreover, the transformer generates much smoother outputs than the nonlinear solver. We provide an analysis of the stability and robustness of the overall closed-loop system. Simulation and experiments are also presented to demonstrate the effectiveness of the proposed method.

Obstacle Avoidance | *Avoid Obstacle Fast with RL-Driven Monocular Depth Estimation*

May 2025 – Now

- Autonomous drone racing is a challenging benchmark for pushing the limits of learning, perception, planning, and control. While expert pilots can navigate tracks using only a single onboard camera by directly mapping visual feedback to control commands, recent pixel-to-control approaches in autonomous racing either rely on simplified intermediate representations or require extensive imitation learning. We propose an end-to-end reinforcement learning framework that enables high-speed obstacle avoidance in complex environments by applying Depth-Anything to estimate depth from monocular RGB input, advancing vision-based autonomous flight and highlighting end-to-end obstacle-avoidance control as a promising direction for real-world robotics.

MemoryVLA | *Memory VLA operations under lifelong memory sequences*

Jun 2026 – Now

- The topic focuses on how robots can use historical experience to make decisions in long-horizon tasks. Since robotic manipulation tasks are often non-Markovian, current observations cannot fully represent the task state. Therefore, VLA models need to combine short-term temporal context with long-term memory. Short-term memory helps the model understand the current task progress, while long-term memory stores and retrieves past experiences, thereby improving the robot's stability, generalization ability, and continuous operation capability in multi-step, long-horizon tasks.